Communication Systems

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INTRODUCTION

With the diversification of society there arose the need of individuals to communicate with their relatives, friends or associates at distant points around the globe. The science of communication involving long distances is called telecommunication. The term communication signifies transmission, reception and processing of information by means of electrical devices.

BASIC CONSTITUENTS OF COMMUNICATION SYSTEM

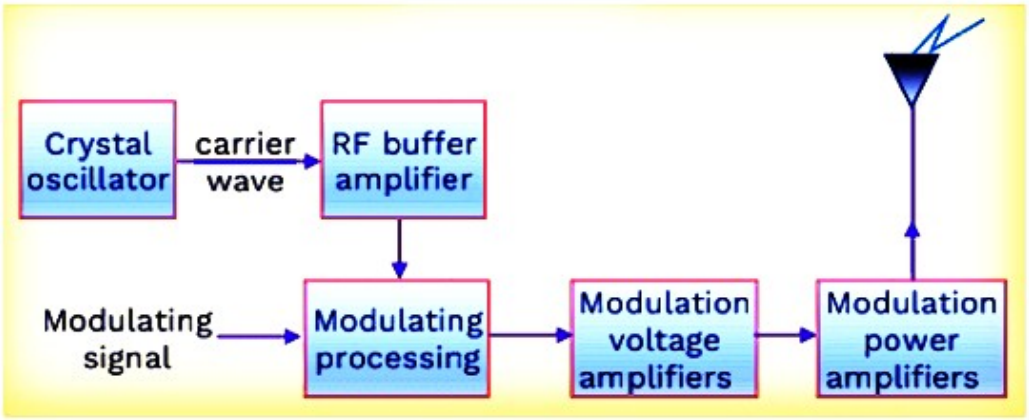
(a) Information:

Useful part of massage that needs to be communicated from source to destination is called as information. The amount of information in a message is expressed in "bits".

(b) Transmitter:

Transmitter is a device that sends electrical signal over a channel by converting it into a suitable form. The information in the incoming signal is being first converted into electrical variations, for example a sound signal must be convert into an electrical signal. In a transmitter this signal is first converted into a suitable form (by modulation, amplification, encoding etc) before transmission over a channel.

Block Diagram of a typical transmitter:



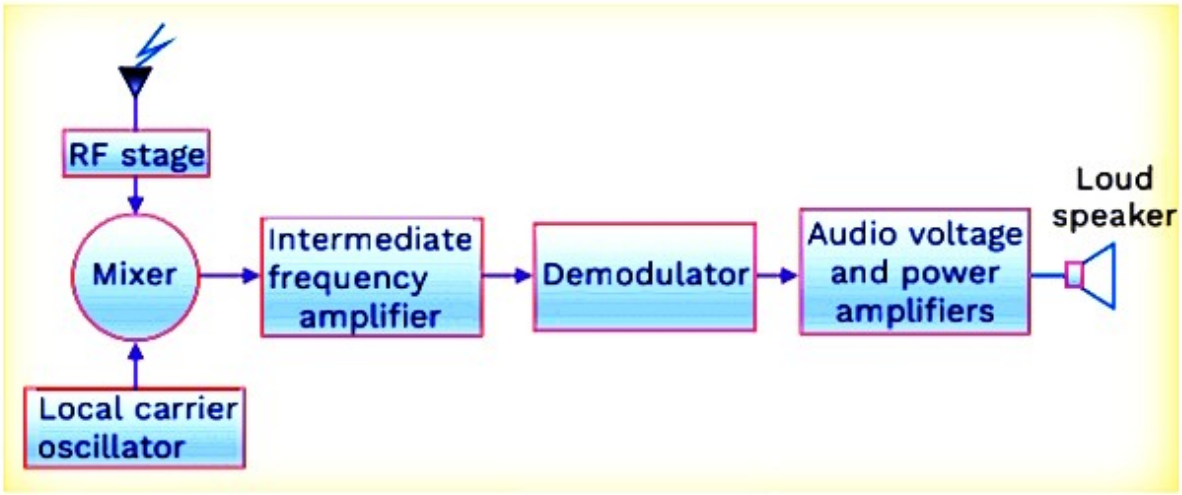
(c) Channel and noise source:

Channel is a physical medium where electrical signal is transmitted by the transmitter. It may be in the form of wires (cables) or may be wireless where it is transmitted over space. When this electrical signal traverse through this channel it get distorted because of imperfection in the medium (Non Linear system). This disrupted unwanted signal is generally of random nature is called noise.

(d) Receiver:

Receiver is a device that receive electrical signal from channel and reconstruct to its recognisable form of source signal. Receiver filter out noise from received electrical signal and make it suitable (by demodulation, decoding etc) for user of information. Output from a receiver can be sent to a high volume video display unit, teleprinter, TV picture tube, pen recorder or computer. The transceiver must be in sync with the modulation and encoding methods used. The figure shows the block diagram of an AM receiver.

Block Diagram of a receiver



BASIC TERMINOLOGY USED IN ELECTRONIC COMMUNICATION SYSTEMS

In order to understand the principles of communication, we should be aware of basic terms that are commonly used.

(i) **Transducer:** An electrical transducer is an instrument that converts input physical variable like displacement, force, temperature, pressure, etc. into corresponding variations in the electrical signal at the output that can be processed with the help of electrical devices. for example- microphones, tape heads, photocells, LEDs, etc.

(ii) **Signal:** Signal is part of information that is converted to electrical form which is suitable for transmission. It is a single values function of time.

(iii) **Attenuation:** It is defined as the loss in the strength (amplitude) of a signal while propagating through a medium/channel.

(iv) **Amplification:** Amplification is the process of increasing strength (amplitude) of a signal with the help of electronic circuit is called as amplifier. Amplification compensate for lost in strength of signal during transmission.

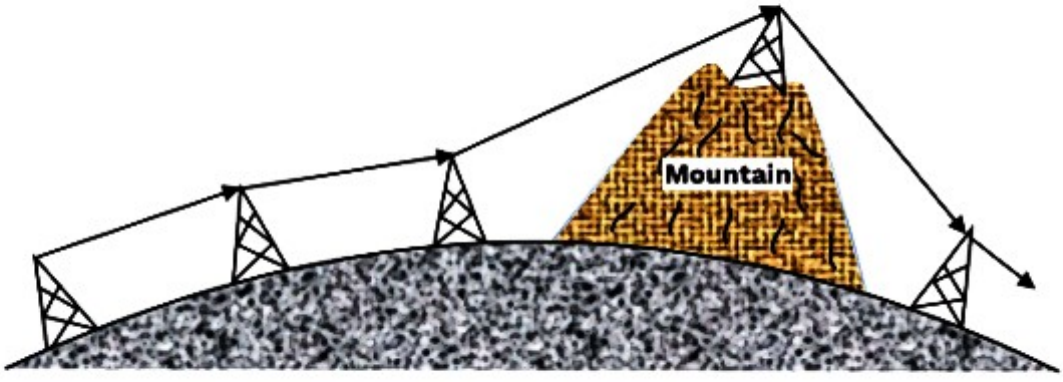
(v) **Range:** It is the longest distance between a destination and a source up to which sufficient strength of the signal is received.

(vi) **Bandwidth**: It is the range of frequency over which any equipment operates without appreciable depreciation in strength of signal, or it can be defined as the amount of spectrum occupied by the signal having appreciable amount of power.

(vii) **Modulation:** The process of transmitting low frequency signal over a high frequency carrier wave by modifying its amplitude frequency, or phase according to masses signal is known as modulation.

(viii) **Demodulation:** It is the process of extracting low frequency masses signal from high frequency carrier wave. It is simply reverse of modulation process.

(ix) **Repeater:** A repeater is a device that receives signal from a transmitter, amplifies it, and retransmits it to the receiver, often with a carrier frequency modification. It is used to increase the range of a communication system and also eliminate noise (distortion) suffered by masses signal during transmission.



Important information:

The unwanted effects in the process of signal transmission are

(a) **Attenuation:** loss in strength (amplitude) of signal during its propagation due to loss in energy.

(b) **Distortion:** disturbance in waveform of message signal.

(c) **Interference:** overlapping of external signals with the message signal.

(d) **Noise:** unwanted electrical signal.

BANDWIDTH OF SIGNALS

Pictures, music, voice, computer data, etc., can be used as message signals. Any massage signal can be written as a combination of sinusoidal function of many frequency with the help of Fourier series expansion. Bandwidth is range of frequency containing appreciable amount of power of masses signal.

For speech signals used for commercial telephonic communication appreciable amount of information is limited in a bandwidth of 2800 Hz (300 Hz–3100 Hz). Bandwidth of music produced by musical instruments is much more than that of speech signal so in order to maintain of quality of music bandwidth of 20 KHz is required during transmission.

Picture signals requires about 4.2 MHz of bandwidth for its transmission. A T.V. signals contains both picture and voice is usually allocated 6 MHz of band width for its transmission.

The transmission medium, also known as a communication channel, is a link that allows information/message signals to travel from source to destination without being distorted. It's a network of electronic highways where signals traverses.

Transmission media have been divided into two types:

1. **Guided transmission medium:** The communication medium or channel used in point-to-point communication between a single transmitter and a single receiver is known as a guided transmission medium. Guided transmission media include, for example, parallel wire lines, twisted pair, and co-axial cable. Other directed transmission mediums include optical fibres. In line communication, a guided transmission medium is used.
2. **Unguided transmission medium:** When there is no point to point contact between the transmission and receiver, communication medium used in unguided transmission medium.

   A single transmitter is accompanied by a huge number of receivers. Unguided transmission mediums include things like free space. It's utilised in radio and television, as well as in space and satellite communication.

   The quality and characteristics of transmission medium depends on

   (i) Nature of signal

   (ii) Nature of transmission medium

BAND WIDTH OF TRANSMISSION MEDIUM

Now has we know that any random signal can be considered as combination of sinusoidal function of different frequencies. Resistance offered to these different frequency would be different. The range of frequency that are appreciably allowed to pass through a channel or medium is known as the **bandwidth** of that medium. widely used Co-axial cable offers a bandwidth of 750 MHz; these cables are generally operated below 18GHz.

Communication through free space using radio waves occurs at frequencies ranging from 10^5 Hz to 10^9 Hz. This range is subdivided further and allocated for various services.

The optical fibre offers very high bandwidth is used in the frequency range of 1 THz to 1000 THz (microwaves to ultraviolet). These fibre can offer a transmission band width more than 100 GHz.

Table: Some important wireless communication frequency bands,

Service	Frequency bands	Comments
Standard AM broadcast	540-1600 kHz	
FM broadcast	88-108 MHz	
Television	54-72 MHz 76-88 MHz 174-216 MHz 420-890 MHz	VHF (very high frequencies) TV UHF (ultra-high frequencies) TV
Cellular Mobile Radio	896-901 MH 840-935 MHz	Mobile to base station Base station to mobile
Satellite Communication	5.925-6.425 GHz 3.7-4.2 GHz	Uplink Downlink

SIGNALS

We can define a message signal as a single-valued time function that delivers data. At any point of time, such a function has a unique value. To make the message signals acceptable for transmission across the channel, they must be transformed into electrical signals.

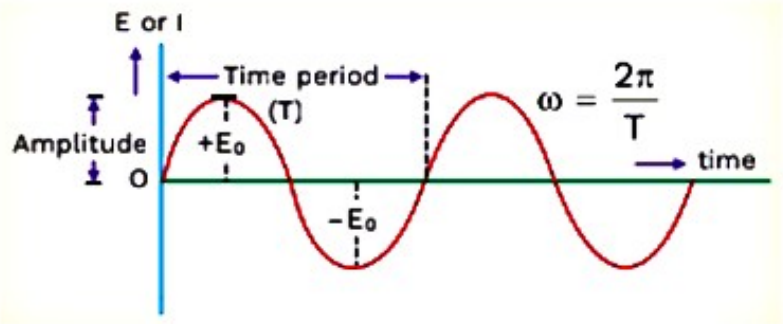
As explained already, we use transducers for the conversion.

The electrical signals can we classified as :

1. **Analog signals:** A signal in which current or voltage value varies regularly with time as a continuous function is called Analog signal.

In the simplest form of an analog signals, amplitude of the signal varies sinusoidally with time, as shown in figure. It is represented by the equation.

$$E = E_0 \sin(\omega t + \phi)$$



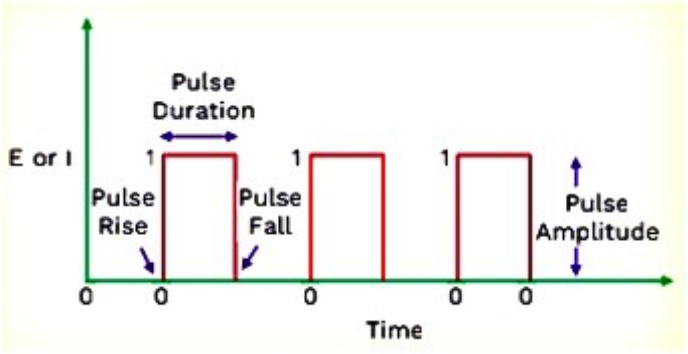
where E_0 is the greatest value of voltage, also known as the amplitude, T is the signal's time period, and $\omega = \frac{2\pi}{T}$ is the signal's angular frequency, and ϕ indicates the phase angle. At different times, such signals can have a variety of values, but they must stay under the range of a maximum value (+ E_0) and a minimum value ($-E_0$).

Speech, music, sound created by a vibrating tuning fork, and fluctuations in light intensity from a picture are all examples of analogue signals. As previously stated, these are transformed into current/voltage variations using appropriate transducers. The frequency range for spoken signals is 300 Hz to 3100 Hz. Speech signals have a band width of 3100 Hz–300 Hz = 2800 Hz. As the frequencies produced by musical instruments are high, therefore, approximate band width for music is 20 kHz.

For transmission of pictures, video signals require a band width of 4.2 MHz. T.V. signals contain both, voice/music, and picture signals. Therefore, band width allocated for transmission of TV. signals is 6 MHz.

Digital signals: In contrast to an analogue signal, a digital signal is a discontinuous function of time. when the value of the current or voltage varies constantly throughout time.

Figure depicts a typical digital signal. Pulses are the most common form of such a signal. Each pulse has two current or voltage levels, denoted by the numbers 0 and 1. A digital signal with a value of zero (0) represents an open circuit, while a digital signal with a value of one (1) represents a closed circuit. The number zero (0) is also known as 'No' or 'Space,' while the number one (1) is known as 'Yes' or 'Mark.' The numbers 0 and 1 are both referred to as bits.



The pulse amplitude, pulse duration or pulse breadth, and pulse position, which represents the time of rise and time of fall of the pulse amplitude, are the important features of a digital signal, as illustrated in figure. Letters printed in a book, (ii) a listing of any data, (iii) the output of a digital computer, (iv) electronic transfer of a document to a remote location through telephone line, i.e. Fax, are all examples of digital signals.

Note that for computer data, the bandwidth allocated is about 600 MHz.

An analog signal can be converted into a digital signal and vice-versa. There are multiple coding schemes useful for digital communication. For example:

(i) Binary coded decimal (BCD) in which a digit is represented as combination of two binary bits (0 or 1).

(ii) American standard code for information interchange (ASCII) is used to represent numbers, letters and certain characters.

PROPAGATION OF ELECTROMAGNETIC WAVES

In radio communication, an antenna at the transmitter produces electromagnetic waves that travel through space and reach the receiving antenna at the other end. The propagation of em waves and the path taken are influenced by a number of factors.

Important information:

The energy of a photon of em wave is calculated by formula $E = \frac{hc}{\lambda}$, where h= 6.6×10^{-34}Js, C = $3 \times 10_{8}$ ms^{-1}. Taking λ for visible light as 5000 Å (on average), 5 Å for X–rays (on average) and 0.005 (on average), the value of X-rays photon energy is keV, visible light photon energy = eV and γ-rays photon energy = MeV.

Ground Wave

To radiate signals with great efficiency, the antennas should be the comparable to the signal wavelength (at least $\lambda/4$). The antennas have a huge physical dimension and are situated on or very close to the ground at longer wavelengths (lower frequencies). Ground-based vertical towers are commonly utilised as transmitting antennas in regular AM broadcasting. The ground has a significant impact on the signal propagation for such antennas. Surface wave propagation is the method of propagation, and the wave glides across the earth's surface.

A wave creates current in the ground which is absorbed by earth surface as it passes over. With increasing frequency, the attenuation of surface waves increases very quickly. The maximum range depends on the power and frequency of transmission (less than a few MHz).

Sky waves

Reflection of radio signals from ionosphere back to earth is used for long distance communication. Short wave (having small wavelength) broadcasting services use sky wave propagation.

The absorption of high-energy light from the sun by air molecules causes ionisation and because of presence of ions this layer is known as ionosphere. The ionosphere is divided further into layers. The degree of lionisation varies depending on the solar radiation and number of molecules available at that region. At higher altitude ionisation is less because of presence of less number of molecules and at lower altitude it is less availability of solar radiation. Ionisation peak occurs at some intermediate hight. For a particular range of frequencies, the ionospheric layer works as a reflector (3 to 30 MHz). Electromagnetic waves with frequency greater than 30 MHz penetrate and exit the ionosphere. The figure depicts these phenomena. Total internal reflection in optics is analogous to the phenomena of bending em waves so that they are deflected towards the earth.

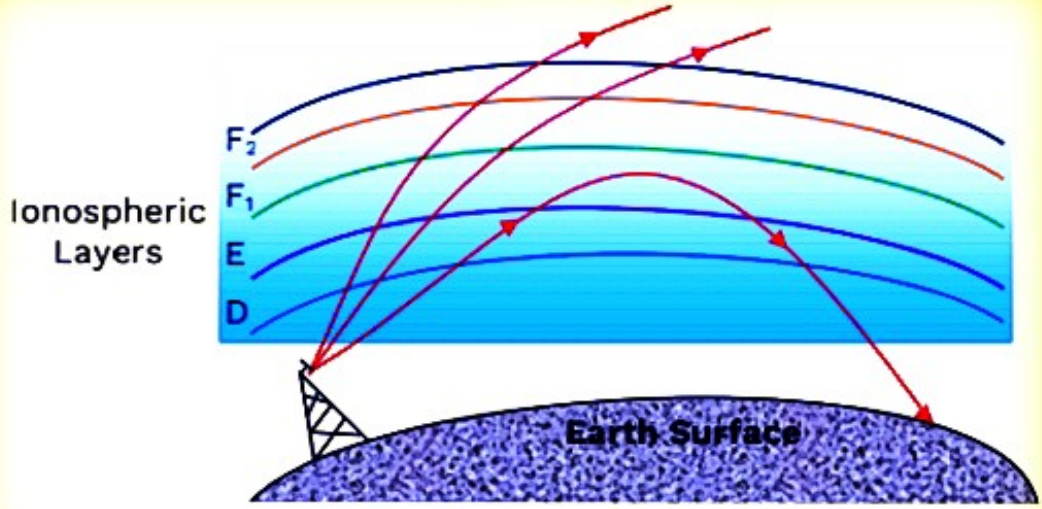


Fig.: Sky wave propagation

Space wave

For frequencies above 40 MHz ionospheric reflection can not be used as these wave travels in a straight path. Line of sight (LOS) communication is used for space wave. The antennas are smaller at these frequencies and can be positioned at heights of many wavelengths above the earth. Direct waves are obstructed at some point by the curvature of the globe due to the line-of-sight aspect of propagation, as seen in figure. If the signal has to be received through the receiving antenna beyond the horizon, it must be high enough to intercept line-of-sight waves.

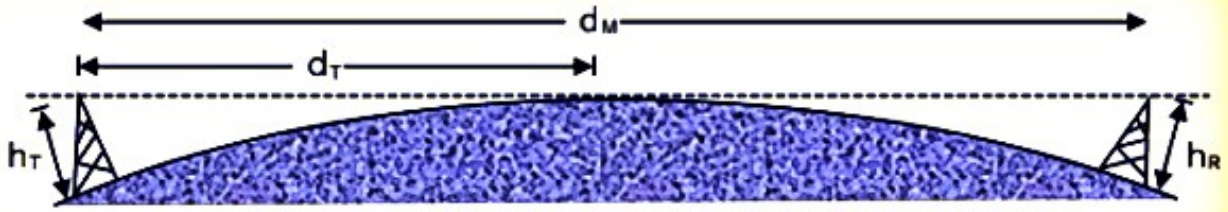


Fig.: Line of sight communication by space waves

The longest line-of-sight distance d_M between the two antennas with heights h_T and h_R above the ground is given by with reference to figure.

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

(We will see derivation of this formulae in the next topic).

Communication methods that utilise the space wave mode of transmission include television broadcast, microwave connectivity, and satellite communication.

Figure summarises the various modes of wave propagation discussed so far.

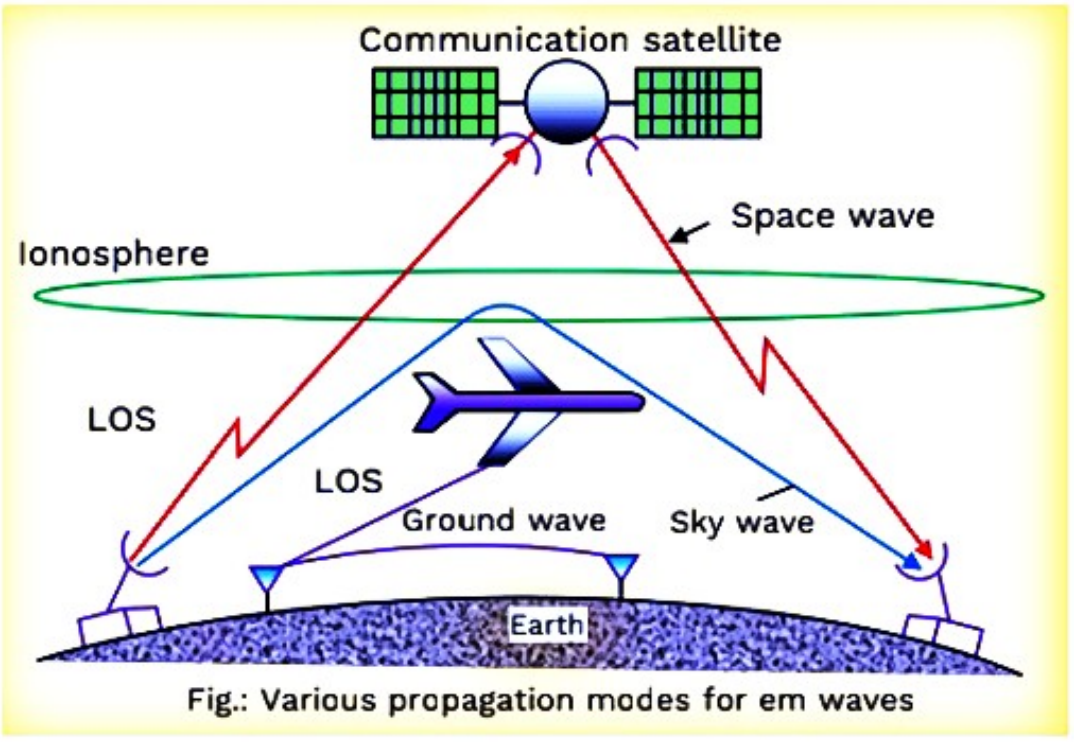


Fig.: Various propagation modes for em waves

Determination of Range:

The height of the transmitting antenna determines the range. Geometrical considerations can readily calculate the AP or PB range. Assume the tower's height is h and the earth's radius is r (OA = OB = OP = t). In the right-angled triangle OOA, we have

$$OQ^2 = QA^2 + OA^2$$

$\because$ $QA = d$

Therefore,

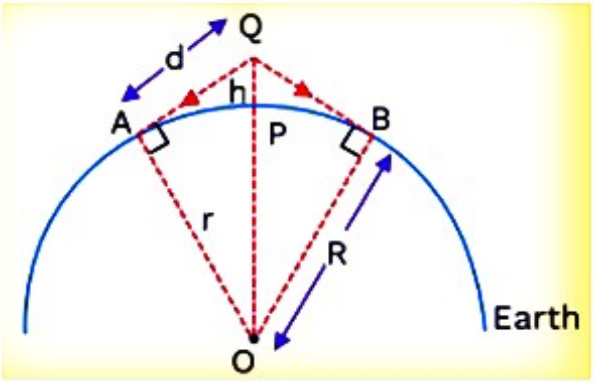
$$(R + h)^2 = R^2 + d^2$$

$$R^2 + h^2 + 2Rh = R^2 + d^2$$

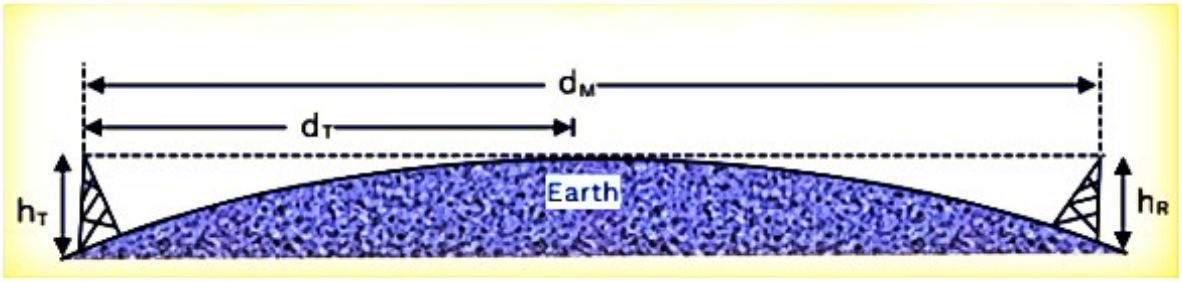
$\because$ $R^2 >> h^2$

$\therefore$ $d^2 = 2Rh$

$$d = \sqrt{2Rh}$$



Hence maximum line-of-sight distance between two antennas will be calculated as follows.



$$d_M = \sqrt{2R \cdot h_T} + \sqrt{2R \cdot h_R}$$

where d_M is the maximum line-of-sight distance between the two antennas

d_T is the height of transmitting antenna

h_R is the height of receiving antenna

radio horizon of transmitting antenna

Antenna

In a communication system, an antenna is critical. It's utilised for both transmitting and receiving radio frequency signals.

An antenna, in reality, is a structure capable of radiating or receiving electromagnetic waves, depending on the situation. An antenna is a metallic item, usually a wire or a group of wires, that is used to convert high frequency electricity into electromagnetic waves and vice versa. A transmitting antenna turns electrical energy (high frequency current) into electromagnetic waves, whereas a receiving antenna converts electromagnetic waves back into electrical energy (high frequency current). Apart from their different functions. transmitting and receiving antennas behave identically.

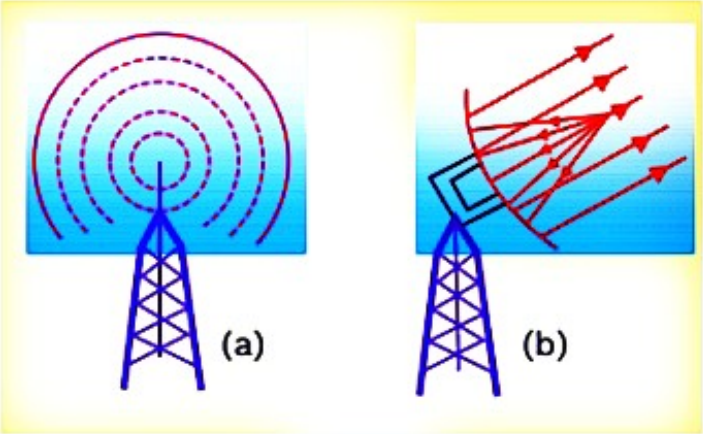
The electromagnetic waves produced when a transmitting antenna is held vertically are polarised vertically. The em waves produced when the same antenna is held horizontally are polarised horizontally.

A Hertz antenna is a straight conductor with a length of half the wavelength of the radio signals to be sent or received ($l = \lambda/2$). This antenna isn't connected to the ground.

A Marconi antenna is a straight conductor with a length of a quarter of the wavelength of the radio signals to be sent or received, i.e. $l = \lambda/4$. It's held vertically, with the bottom end resting on the ground. The voltage and current distributions set up in the antenna are reflected by the ground. The electromagnetic waves released by the ground system of the (Marconi) antenna are identical to those emitted by a Hertz antenna that is not grounded.

The frequency of the carrier wave and the directivity of the beam, among other factors, influence antenna design. Antennas come in two varieties:

1. **Dipole Antenna,** shown in fig.(a) is used in transmission of radio waves. It is omni directional.



2. **Dish type Antenna,** A directional antenna is depicted in figure (b). At the focus of the reflector is a parabolic reflector with an active element called a dipole or horn feed. Waves can be transmitted in a specific direction using a dish antenna. It can also only receive waves that are directed towards it. The signal is delivered to the active element, which directs it to the reflector for transmission. As seen in figure (1), the signals are then transferred in the form of a parallel beam (b). The waves directed at the dish are reflected onto the active element, which turns them into electrical impulses for reception. Radar and satellite communication frequently employ dish antennas.

1. **Size of the antenna or aerial:** Both transmission and reception require an antenna or aerial. Each antenna should be comparable to the size of wavelength of signals (at least $\lambda/4$ in size), so that, the antenna can detect the signal's time fluctuation.

   For a speech signal having frequency $\nu = 15$ kHz, the wavelength,

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{15 \times 10^3} = 20000\text{m}$$

   The required minimum length of the antenna $= \frac{\lambda}{4} = \frac{20000}{4} = 5000$ metre. This is practically impossible to setup an operate antenna of this size. If transmission frequency is raised to 1 MHz, then

Then required minimum length of antenna, $\frac{\lambda}{4} = \frac{c}{4v} = \frac{3 \times 10^8}{4 \times 10^6} = 75\,m$

This length of antenna is practically possible to setup. Therefore, there is an urgent need of converting the information contained in our original low frequency base-band signal into high or radio frequencies before transmission.

2. **Effective power radiated by antenna.** Theoretical studies reveal that power P radiated from a linear antenna of length l is proportional $(l/\lambda)^2$, i.e., $p \propto \left(\frac{l}{\lambda}\right)^2$

Because high powers are required for good transmission, the wavelength λ should be narrow or the frequency v should be high for a given antenna length.

As a result, this aspect also emphasises the importance of employing high-frequency transmission.

3. **Mixing up of signals from different transmitters.** By using modulation technique many message signals having same bandwidth (Range of frequencies) can be transmitted over a single channel without interfering with each other. This is achieved by transmitting different message signals over carrier waves corresponding to different frequencies. These message signals over different carrier frequencies can be separated by using suitable frequency and then retrieving the message from that carrier wave with the help of demodulation. This is what various radio and television broadcast stations are doing. All the three reasons explained above suggest that there is a need for transmission at high frequencies. This is achieved by a process, called modulation.

As we have already discussed a carrier wave is a high frequency signal that may be :

(i) Continuous (sinusoidal) wave, or

(ii) Pulse which is discontinuous.

Let sinusoidal carrier wave be represented as

$$C_{(t)} = A_c \sin(\omega_c t + \phi)$$

where $C_{(t)}$ is the signal strength (current or voltage) of carrier wave at any time t.

A_c = amplitude of carrier wave

$\omega_c = 2\pi \nu_c$ is angular frequency of carrier wave (in rad/s)

ν_c = frequency of carrier wave in Hz

ϕ = initial phase of the carrier wave

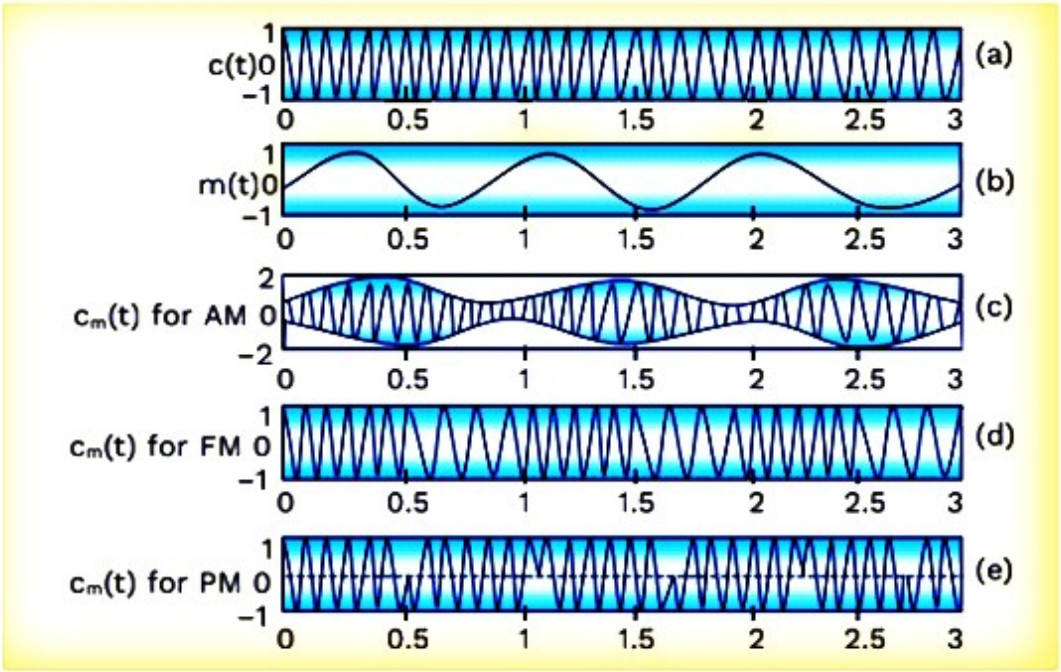
This carrier wave have three independent parameter amplitude (A_c), angular frequency (ω_c) and phase (ϕ). Any of these parameter can be altered in accordance with the message signal [m(t)]. Based on the parameter which is altered modulation is of three types :

(i) Amplitude modulation (AM)

(ii) Frequency modulation (FM)

(iii) Phase modulation (PM)

In figure part (a) shows a sinusoidal carrier wave $C_{(t)}$, part (b) shows a modulating signal m(t); part (c) shows amplitude modulated wave, in which amplitude varies, part (d) shows frequency modulated wave, in which frequency varies and part (e) shows phase modulated wave, in which phase varies in accordance with modulating signal.



Pulse Amplitude, Pulse Duration or Pulse Width, and Pulse Position are three independent parameters in a pulse (representing the time of rise or fall of the pulse amplitude). The modulating baseband (AF) signal can change any of these parameters, resulting in Pulse Amplitude Modulation (PAM), Pulse Duration Modulation (PDM), Pulse Width Modulation (PWM), and Pulse Position Modulation (PPM). We will discuss only amplitude modulation in this chapter.

AMPLITUDE MODULATION

The carrier wave's amplitude linearly varies according to the message signal as

$$A_c = a + b\, m(t)$$

Depending on the values of a and b, there are different type of amplitude modulation but we will discuss only one in this basic level course.

Let us consider the following sinusoidal modulating signal (message signal) :

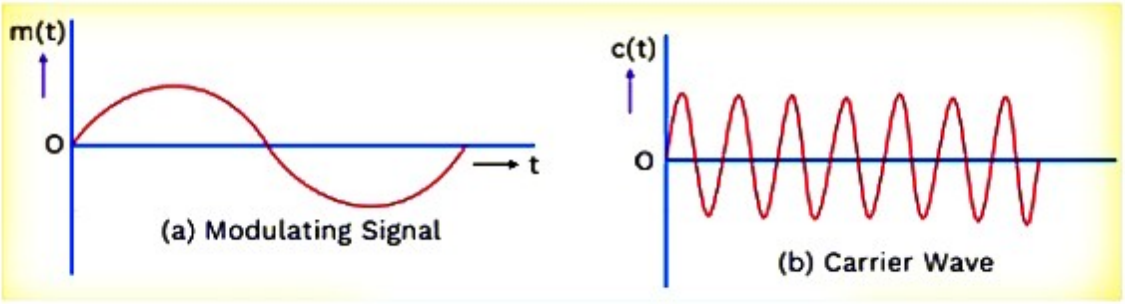
$$m(t) = A_m \sin \omega_m t \qquad \ldots.(1)$$

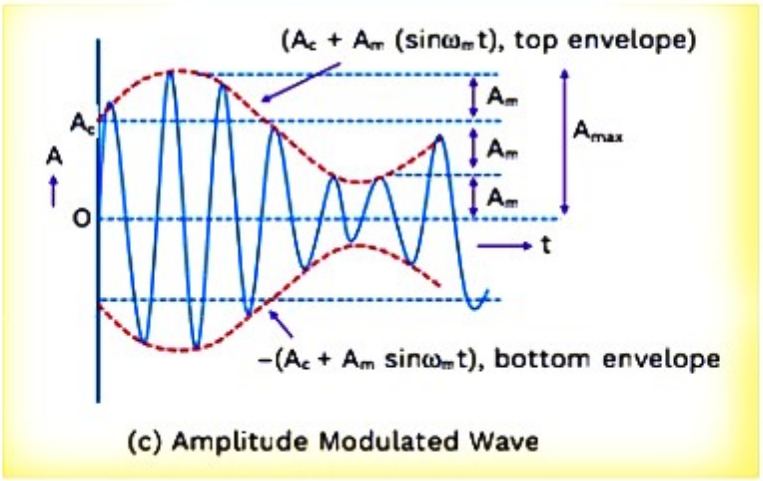
However message signal m(t) is a random signal that is unpredictable as speech signal, but in order to have mathematical simplicity we will assume message signal to be a simple sinusoid of single frequency as represented in figure (a).

Let us consider carrier wave of amplitude Ac and angular frequency $\omega_c = 2\pi \nu_c$ as

$$c(t) = A_c \sin \omega_c t$$

This is represented in fig.(b)





In amplitude modulation, amplitude of carier wave is changed from A_c to $(A_c + A_m \sin \omega_m t)$, therefore, we represent amplitude modulated wave as

$$c_m(t) = \left(A_c + A_m \sin\omega_m t\right)\sin\omega_c t \qquad \text{.....(2)}$$

$$= A_c\left(1 + \frac{A_m}{A_c}\sin\omega_m t\right)\sin\omega_c t$$

Put $\frac{A_m}{A_c} = \mu$

Here, μ is called **amplitude modulation index.**

$$\therefore \quad c_m(t) = A_c \sin\omega_c t + \mu A_c \sin\omega_m t.\sin\omega_c t \qquad \text{....(3)}$$

In practice, the value of μ is kept less than 1 to avoid distortion which can be seen in figure (c).

Using the trigonometric relations,

$$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)].$$ We get from .....(4)

$$c_m(t) = A_c \sin \omega_c t + \frac{\mu A_c}{2} \times \left[\cos\left(\omega_c - \omega_m\right)t - \cos\left(\omega_c + \omega_m\right)t\right] \quad ....(5)$$

This modulated signal consist of a carrier wave in addition to this it also contain sinusoidal wave of frequency $\omega_c + \omega_m$ and other of frequency $\omega_c + \omega_m$. These two additional waves are called side bands.

Frequency of lower side band,

$$\omega_{LSB} = \omega_c - \omega_m \quad ...(6)$$

and frequency of upper side band,

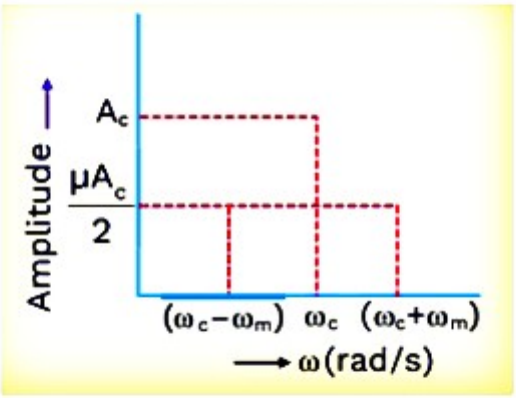
$$\omega_{USB} = \omega_c + \omega_m \quad ....(7)$$

Band width of amplitude modulated wave would be

$$= \omega_{USB} - \omega_{LSB}$$

$$= \left(\omega_c + \omega_m\right) - \left(\omega_c - \omega_m\right) \quad ....(8)$$

$$= 2\omega_m$$



i.e., **Band width of modulated signal = twice the frequency of modulating signal.** The frequency spectrum of amplitude modulated wave is shown in figure. The two side band frequencies have equal amplitude (= $\mu A_c/2$), which never exceeds half the carrier amplitude as μ is less than 1.

Power and Current relation in AM wave

Average power or cycle in the unmodulated carrier wave is.

$$P_c = \frac{A_c^2}{2R} \qquad ...(9)$$

Here, R is resistance (of antenna) in which power is dissipated. It can be shown that total power/cycle in the modulated wave is.

$$P_t = \underbrace{\frac{A_c^2}{2R}}_{\text{carrier power}} + \underbrace{\frac{(\mu A_c / 2)^2}{2R} + \frac{(\mu A_c / 2)^2}{2R}}_{\text{side band power}}$$

$$P_t = \frac{A_c^2}{2R} + \underbrace{\frac{(\mu A_c)^2}{4R}}_{\text{side band power}}$$

$$P_t = P_c\left(1 + \frac{\mu^2}{2}\right) \qquad ....(10)$$

$$\therefore \quad \frac{P_t}{P_c} = 1 + \frac{\mu^2}{2}$$

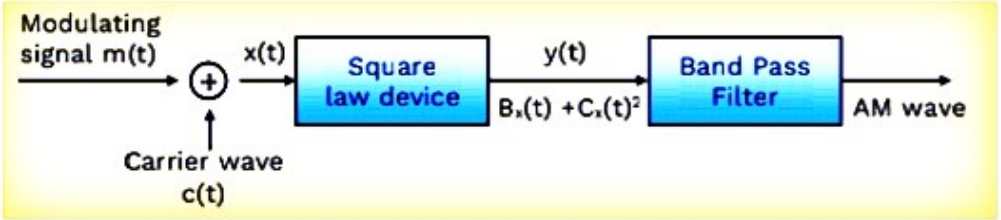
If I_t and I_c be the r.m.s. current corresponding to modulated signal and unmodulated signal respectively, then

$$P_t = I_t^2 R \quad \text{and} \quad P_c = I_c^2 R$$

$$\therefore \quad \frac{P_t}{P_c} = \frac{I_t^2}{I_c^2} = \left(1 + \frac{\mu^2}{2}\right) \text{ or } \frac{I_t}{I_c} = \sqrt{1 + \frac{\mu^2}{2}} \qquad ....(11)$$

PRODUCTION OF AMPLITUDE MODULATED WAVE

One of the simplest methods for producing amplitude modulated wave is by using square law device as shown in block diagram given below.



Let us assume :

Message signal, $m(t) = A_m \sin \omega_m t$

Carrier wave, $c(t) = A_c \sin \omega_m t$

When the modulating signal is added to the carrier wave, the signal produced would be

$$x(t) = A_m \sin\omega_m t + A_c \sin\omega_c t \quad \text{.....(12)}$$

When this signal is passed through a **square law device**, which is a **non linear device**, it produces an output.

$$y(t) = Bx(t) + C[x(t)]^2 \quad \text{....(13)}$$

Here, B and C are arbitrary constants. Using equation (12), we get

$$y(t) = B\left[A_m \sin\omega_m t + A_c \sin\omega_c t\right] + C\left[A_m \sin\omega_m t + A_c \sin\omega_c t\right]^2$$

$$= B\left[A_m \sin\omega_m t + A_c \sin\omega_c t\right] + C\left[A_m^2 \sin^2\omega_m t + A_c^2 \sin^2\omega_c t + 2A_m A_c \sin\omega_m t \sin\omega_c t\right]$$

Using the trigonometric relations

$$\sin^2 A = \frac{1-\cos 2A}{2}$$

$$\text{And } \sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$$

$$\text{We get } y(t) = BA_m \sin\omega_m t + BA_c \sin\omega_c t + \frac{C}{2}\left(A_m^2 + A_c^2\right) - \frac{C}{2}A_m^2 \cos 2\omega_m t$$

$$-\frac{C}{2}A_c^2 \cos 2\omega t + CA_m A_c \cos\left(\omega_c - \omega_m\right)t - CA_m A_c \cos\left(\omega_c + \omega_m\right)t$$

This equation contains a d.c. term $\frac{C}{2}\left(A_m^2 + A_c^2\right)$ and sinusoidal waves of frequency $2\omega_m$; $2\omega_m, \omega_c, 2\omega_c$; $\left(\omega_c - \omega_m\right)$ and $\left(\omega_c + \omega_m\right)$. This signal is passed through a band pass filter which allow only the term of sinusoid having angular frequency ω_c, $\omega_c + \omega_m$ and $\omega_c - \omega_m$. Other term which are not required would be rejected by the filter.

This amplitude modulated signal is first amplified by the power the amplifier, so that it is not easily corrupted by low power noise present in a channel. After amplification it is fed to an antenna of appropriate size for radiation. The block diagram of a transmitter is shown in figure (a)

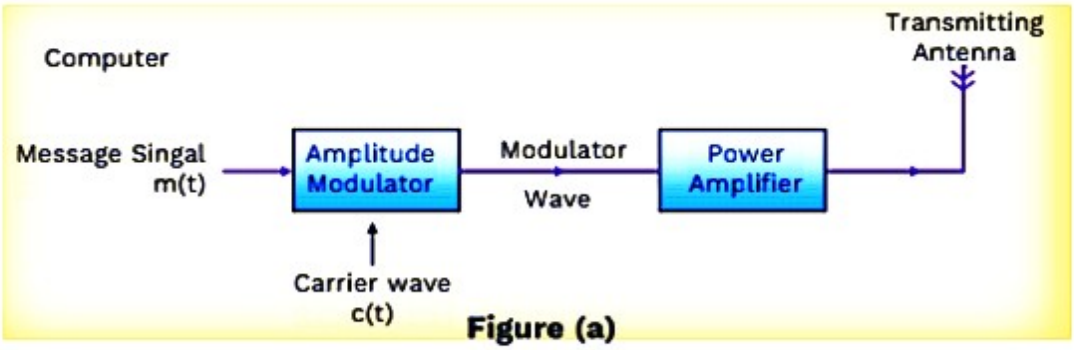


Figure (a)

DETECTION OF AMPLITUDE MODULATED

Demodulation is the process of recovering original low frequency signal from high frequency carrier wave.

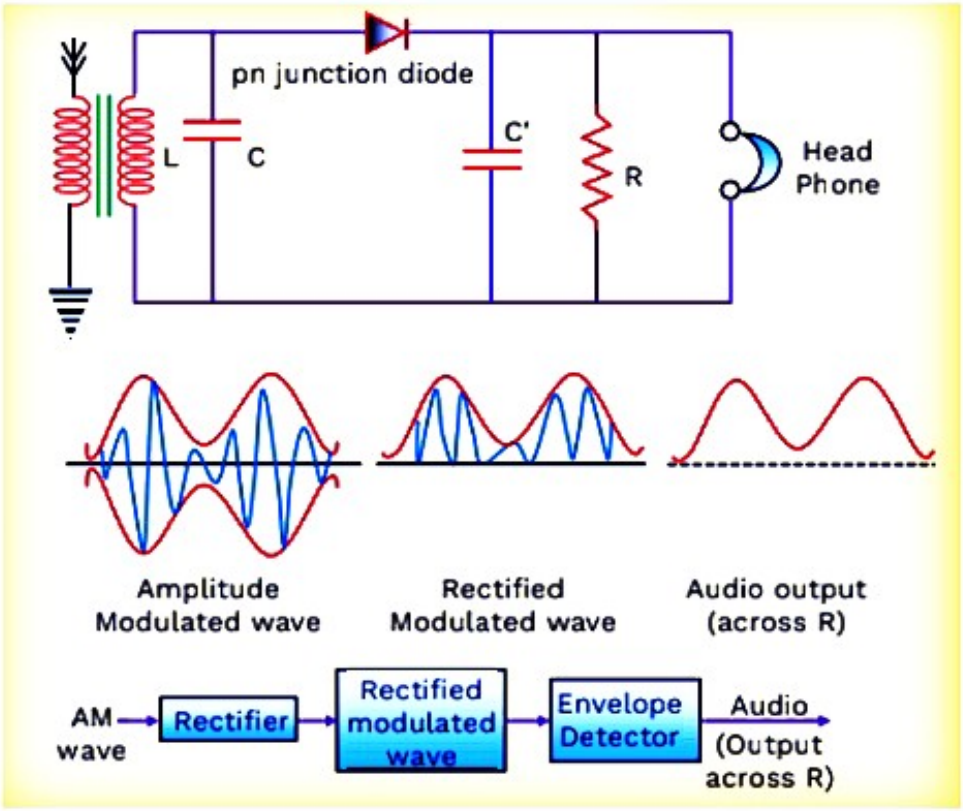
Thus, a demodulator performs the following functions :

(a) Rejecting unwanted signal (frequency) and selecting only those frequencies corresponding to desired message signal that is upper side band and lower side band frequencies.

(b) Amplifying and demodulating the desired signal coming from band passed signal (here LC tuned circuit).

(c) Presenting a message signal in physical form required to the end user (voice, picture etc.)

The input circuit consists of a tuned circuit (Band pass filter) comprising an inductor L and a variable capacitor C. This allow only desired signal from the different signal coming to the antenna. After this band pass filter signal goes to envelop detector which detect only positive half cycle which is our message signal m(t).



Envelope Detector

Envelope detector contains a junction diode and a parallel RC circuit, as shown in the diagram. Here junction diode only allows positive part of signal to pass through it. The charging of the capacitor depends on the magnitude of positive peaks. It also discharges as the time passes, in order to trace peaks having less magnitude. In order to trace the envelope of amplitude modulated signal which is our message signal, time constant RC of this circuit must be much greater than time period of high frequency carrier wave,

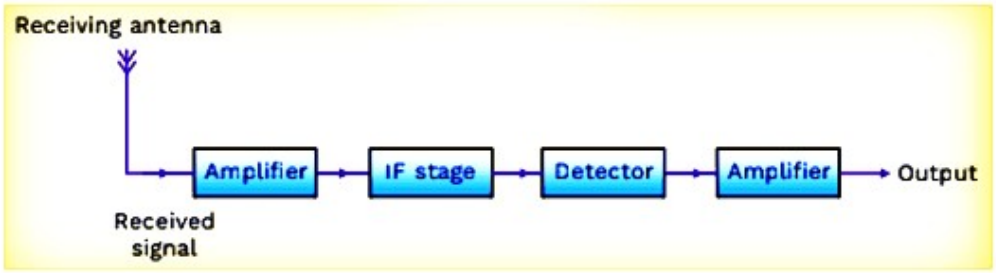
$$\text{i.e., } RC >> T_c \text{ or } \frac{1}{\nu} << RC$$

Also, in order to trace all the peaks time constant RC must be much less than frequency of a message signal,

$$\text{i.e., } RC << Tm \text{ or } \frac{1}{\nu_m} >> RC$$

Maximum modulated frequency that could be detected by the envelope detector would be

$$f_m = \frac{1}{2\pi RC\ \mu}$$



Now, the signal coming from envelop detector has wave form similar to message signal. This message signal without noise would be amplified according to the end user of information.